

LDC ERP Meeting — Parts Integration & Store GR Write-Back

Date: 2026-06-24

Purpose: Align with LDC and the ERP consultant on the parts data flow: (1) confirm the parts read endpoint and field mapping, and (2) agree on the mechanism for SM to push quantity updates back to ERP when goods are issued to a requester.

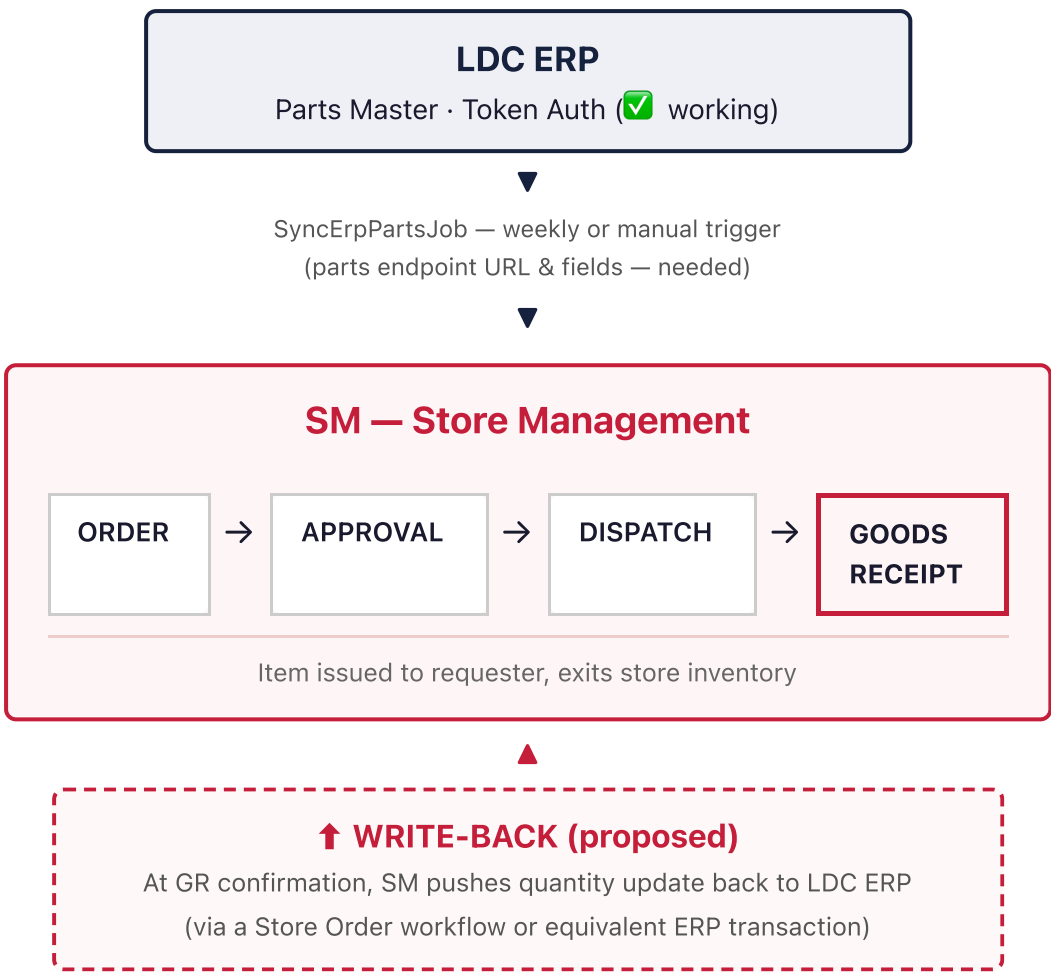
1. Where We Are Today

We have the LDC ERP connection established and working:

Capability	Status
Authentication	✔ Token endpoint working (<code>client_id</code> + <code>client_secret</code> → bearer token)
Asset endpoint	✔ URL confirmed, field mapping agreed
Parts endpoint	✘ Needed — URL, auth (same token?), and list of available fields
ERP → SM sync (parts)	Designed but blocked on parts endpoint details
SM → ERP write-back	Not yet implemented; this is the purpose of the meeting

The read-only parts sync (`SyncErpPartsJob`) is designed and partially built — it will pull parts master data from LDC ERP into SM on a weekly schedule (manual trigger also available). SM then manages the part catalogue, inventory, and the Order → Approval → Dispatch → Goods Receipt workflow. ATMS reads parts from SM for Work Order part-request forms.

2. Current Flow & Where Write-Back Fits



Key points

Aspect	Detail
Read direction	ERP → SM. Parts master data synced on schedule.
Write-back direction	SM → ERP (proposed). Triggered at GR — when item is issued to requester and leaves store.
Current status	Token auth ✅ · Asset endpoint ✅ · Parts endpoint ❌ (blocking)
Write-back mechanism	Not yet designed. Needs ERP input on transaction type (Store Order? Stock Issue?).

3. What We Need from This Meeting

Q1 — What is the Parts endpoint, and what fields are available?

We have the token endpoint and asset endpoint working. We need the same for parts:

- **Parts endpoint URL** — what is it, and does it use the same bearer token?

- **Available fields** — which fields does the ERP expose for each part? For example:
 - Part ID / code
 - Name / description
 - Unit of measure
 - Category
 - Current status (active / inactive?)
 - Stock-on-hand quantity (does ERP expose this, or only SM tracks it?)
 - Last updated timestamp
- **Pagination** — cursor-based? Page size limits?
- **Incremental sync** — is there an `updated_since` filter?

Q2 — Can we push quantity updates back through a Store Order workflow?

This is the core question. When SM issues items at GR, the store's stock decreases. LDC wants the ERP to reflect this. We need to understand the mechanism:

- Does LDC ERP have a **Store Order** or **Stock Issue** transaction type that SM can call to decrement inventory quantities?
- What is the **endpoint and payload** for such a transaction? For example:
 - Part ID / code
 - Quantity issued (decrement)
 - Transaction date
 - Order reference (should SM carry an ERP PO number?)
 - Requester / recipient identifier
 - Store / warehouse code
- Is the **same token** used for write operations, or does it need elevated scopes?
- Is there an existing **API spec or example payload** we can review?

Q3 — How are partial issues and edge cases handled?

Real-world issuance rarely matches the order exactly:

- **Partial issue:** Ordered 10, only 7 available to issue. Can the ERP accept a GR for 7 (partial)?
 - **Split issue:** One order fulfilled across multiple GRs over time. Does ERP support multiple GR transactions against one order?
 - **Cancellation / return:** If issued items are returned, is that a separate reversal transaction or a negative GR?
 - **Idempotency:** If SM retries a push due to a network error, how does the ERP prevent duplicate GR records? Is there an idempotency key?
 - **Reconciliation:** If the push fails, what is the recovery path — manual re-push, or ERP-side adjustment?
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4. Technical Implications (if write-back agreed)

Area	Impact
Parts read sync	Add parts endpoint URL and field mapping to <code>config/erp.php</code> . Confirm token works for parts.
ERP adapter	<code>ErpSource</code> contract gains write methods (<code>postGoodsReceipt()</code> / <code>postStoreOrder()</code>).
SM workflow	GR confirmation triggers a queued job to push to ERP.
Error handling	Queue job with retry/backoff; audit log records success/failure.
Auth	Confirm existing token scopes cover write operations.
Field mapping	Map SM part/order data to ERP's expected Store Order / GR transaction payload.